Year 11 Higher

REVISION PACK

Shape and Space 5

Name: _____

Class: _____

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Worked example: the angle in a semicircle

A, B and C are points on a circle, centre O. AB is a diameter. Angle CAB = 34°. Work out angle ABC, z.
Give a reason for your answer.

Step 1	Angle in a semicircle	$ACB = 90^\circ$
Step 2	Angles in a triangle	$z = 180 - 90 - 34$
Step 3	Answer	$z = 56^\circ$

Practice: the angle in a semicircle

1.1

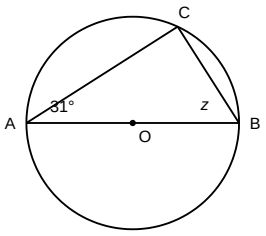


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 31°
Work out the value of z.
Give a reason for your answer.

1.2

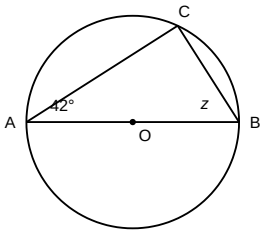


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 42°
Work out the value of z.
Give a reason for your answer.

1.3

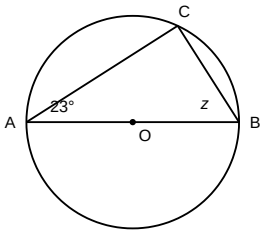


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 23°
Work out the value of z.
Give a reason for your answer.

Practice: the angle in a semicircle (continued)

1.4

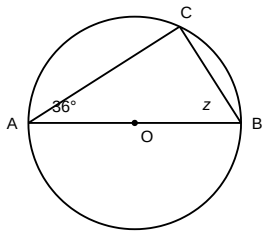


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 36°
Work out the value of z.
Give a reason for your answer.

1.5

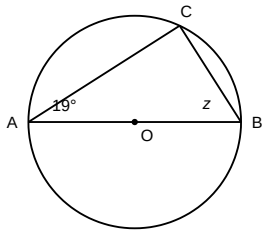


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 19°
Work out the value of z.
Give a reason for your answer.

1.6

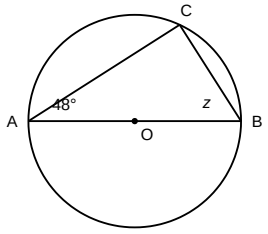


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 48°
Work out the value of z.
Give a reason for your answer.

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Worked example: angles in the same segment

A, B, C and D are points on a circle. Angle ACB = 44°. Work out angle ADB, w. Give a reason for your answer.

Step 1	Same segment: equal	$w = 44^\circ$
Step 2	Reason	angles in the same segment are equal

Practice: angles in the same segment

1.7

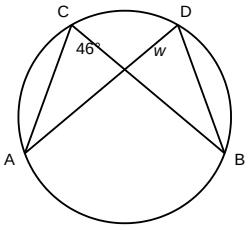


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ACB = 46°
Work out the value of w.
Give a reason for your answer.

1.8

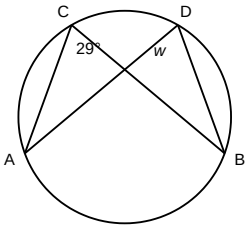


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ACB = 29°
Work out the value of w.
Give a reason for your answer.

1.9

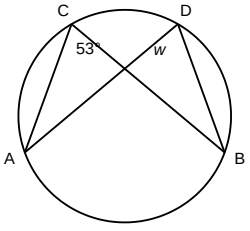
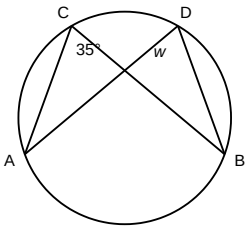


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ACB = 53°
Work out the value of w.
Give a reason for your answer.

Practice: angles in the same segment (continued)

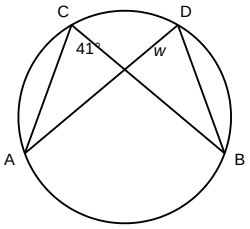
1.10



A, B, C and D are points on a circle.
Angle ACB = 35°
Work out the value of w.
Give a reason for your answer.

Diagram **NOT** accurately drawn

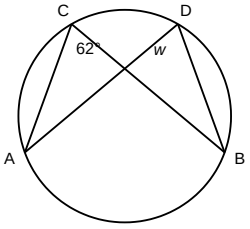
1.11



A, B, C and D are points on a circle.
Angle ACB = 41°
Work out the value of w.
Give a reason for your answer.

Diagram **NOT** accurately drawn

1.12



A, B, C and D are points on a circle.
Angle ACB = 62°
Work out the value of w.
Give a reason for your answer.

Diagram **NOT** accurately drawn

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Worked example: cyclic quadrilaterals

$ABCD$ is a cyclic quadrilateral. Angle ABC = 112°. Work out angle ADC, x. Give a reason for your answer.

Step 1	Opposite angles add to 180°	$x = 180 - 112$
Step 2	Answer	$x = 68^\circ$

Practice: cyclic quadrilaterals

1.13

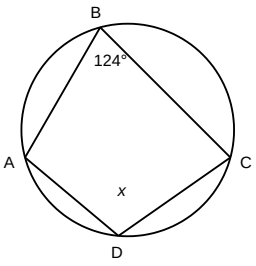


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 124°
Work out the value of x.
Give a reason for your answer.

1.14

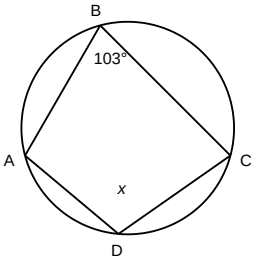


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 103°
Work out the value of x.
Give a reason for your answer.

1.15

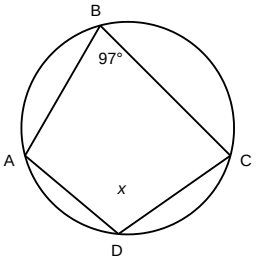


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 97°
Work out the value of x.
Give a reason for your answer.

Practice: cyclic quadrilaterals (continued)

1.16

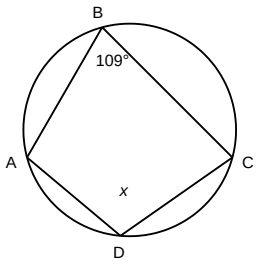


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 109°
Work out the value of x.
Give a reason for your answer.

1.17

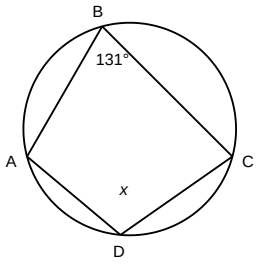


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 131°
Work out the value of x.
Give a reason for your answer.

1.18

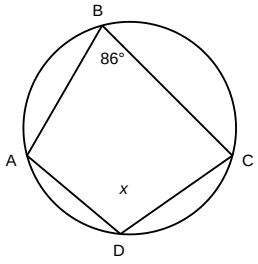


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 86°
Work out the value of x.
Give a reason for your answer.

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Worked example: the alternate segment theorem

TAS is a tangent to a circle at A. Angle BAS = 64°. Work out angle ACB, y. Give a reason for your answer.

Step 1	Tangent and chord	64°
Step 2	Answer	$y = 64^\circ$
Step 3	Reason	alternate segment theorem

Practice: the alternate segment theorem

1.19

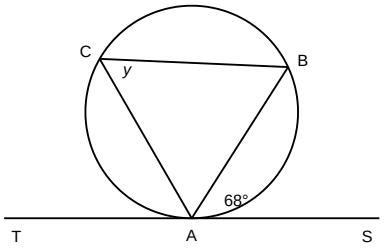


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 68°
Work out the value of y.
Give a reason for your answer.

1.20

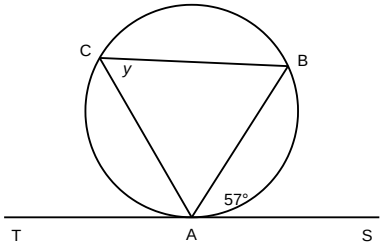


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 57°
Work out the value of y.
Give a reason for your answer.

1.21

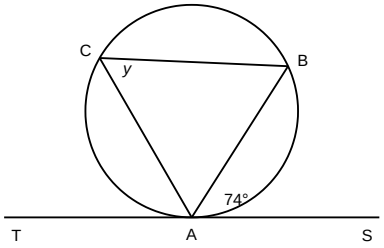


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 74°
Work out the value of y.
Give a reason for your answer.

Practice: the alternate segment theorem (continued)

1.22

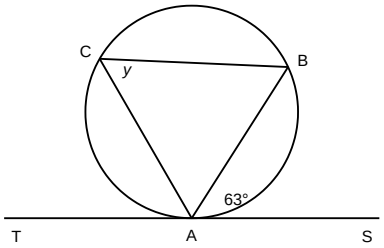


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 63°
Work out the value of y.
Give a reason for your answer.

1.23

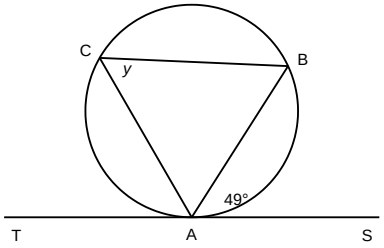


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 49°
Work out the value of y.
Give a reason for your answer.

1.24

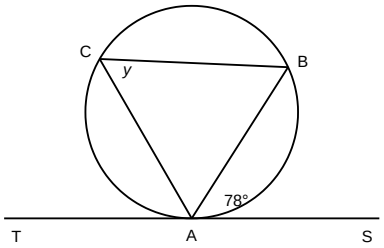


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 78°
Work out the value of y.
Give a reason for your answer.

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Worked example: the angle at the centre

O is the centre of a circle. Angle AOB = 130°. Work out the size of angle ACB. Give a reason for your answer.

Step 1	Centre is twice the edge	$130 \div 2$
Step 2	Answer	65°

Practice: the angle at the centre

1.25

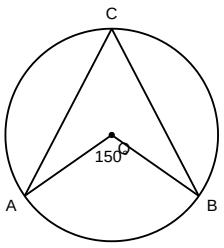


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 150°
Work out the size of angle ACB.
Give a reason for your answer.

1.26

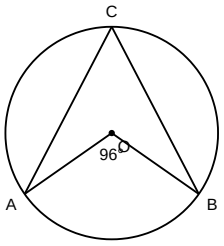


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 96°
Work out the size of angle ACB.
Give a reason for your answer.

1.27

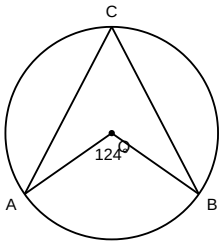


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 124°
Work out the size of angle ACB.
Give a reason for your answer.

Practice: the angle at the centre (continued)

1.28

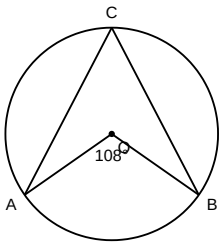


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 108°
Work out the size of angle ACB.
Give a reason for your answer.

1.29

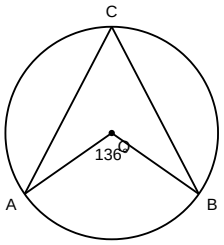


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 136°
Work out the size of angle ACB.
Give a reason for your answer.

1.30

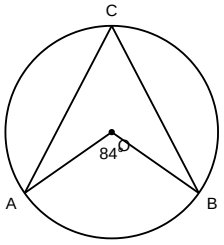


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 84°
Work out the size of angle ACB.
Give a reason for your answer.